UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

BIOLOGY **5090/01**

Paper 1 Multiple Choice **October/November 2009**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

* 5 2 7 5 8 1 4 3 1 4 *

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

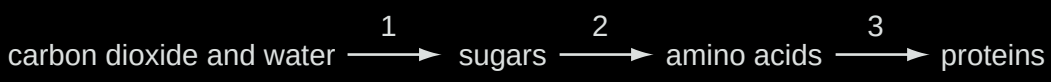
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



What type of cell is this?

	type of cell	reason
A	animal cell	outer layer is the cell membrane
B	bacterium	no chromosomes are visible
C	plant cell	cytoplasm is visible
D	plant cell	cell wall is visible



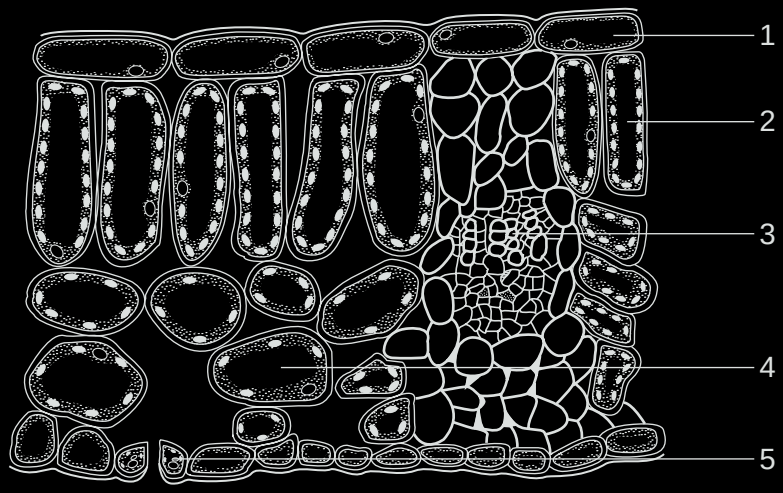
Which stage or stages depend on the use of nitrate ions as a raw material?

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

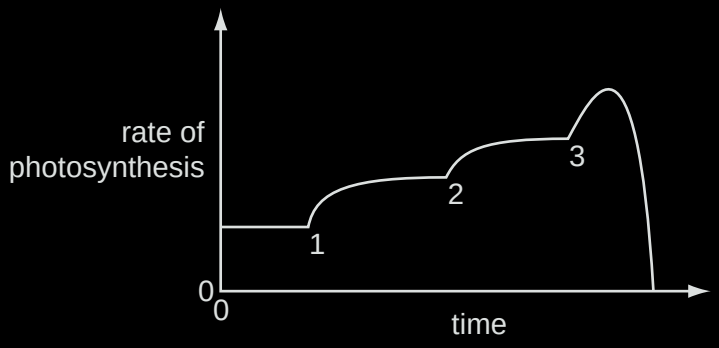


Which cells absorb carbon dioxide?

A 1, 2 and 3 **B** 2, 3 and 4 **C** 2, 4 and 5 **D** 1, 3 and 5

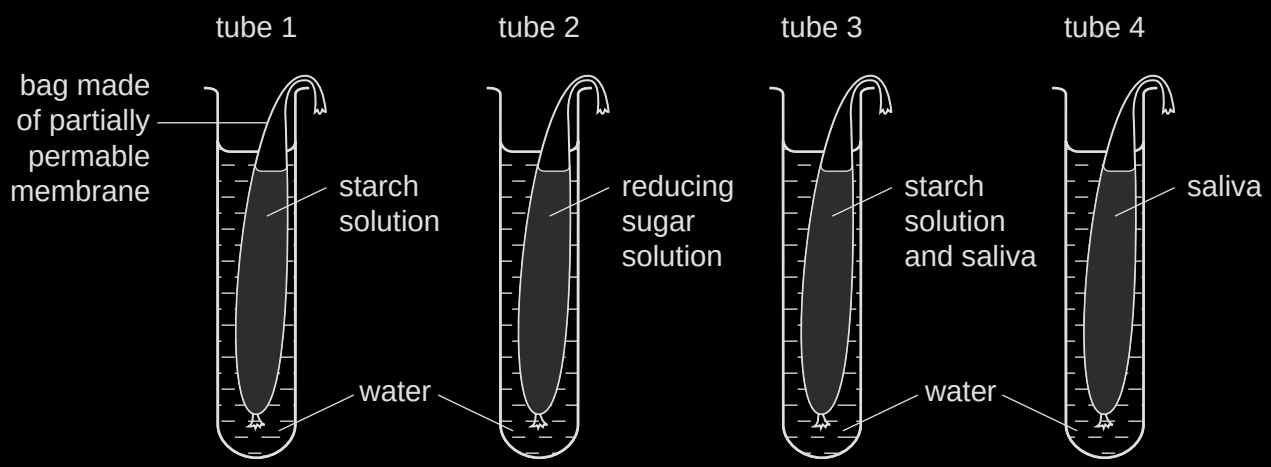
7 Temperature, light intensity and carbon dioxide concentration are three limiting factors in photosynthesis.

In an experiment, each factor is increased in turn. The graph shows the results.



Which numbered points represent when each factor was increased?

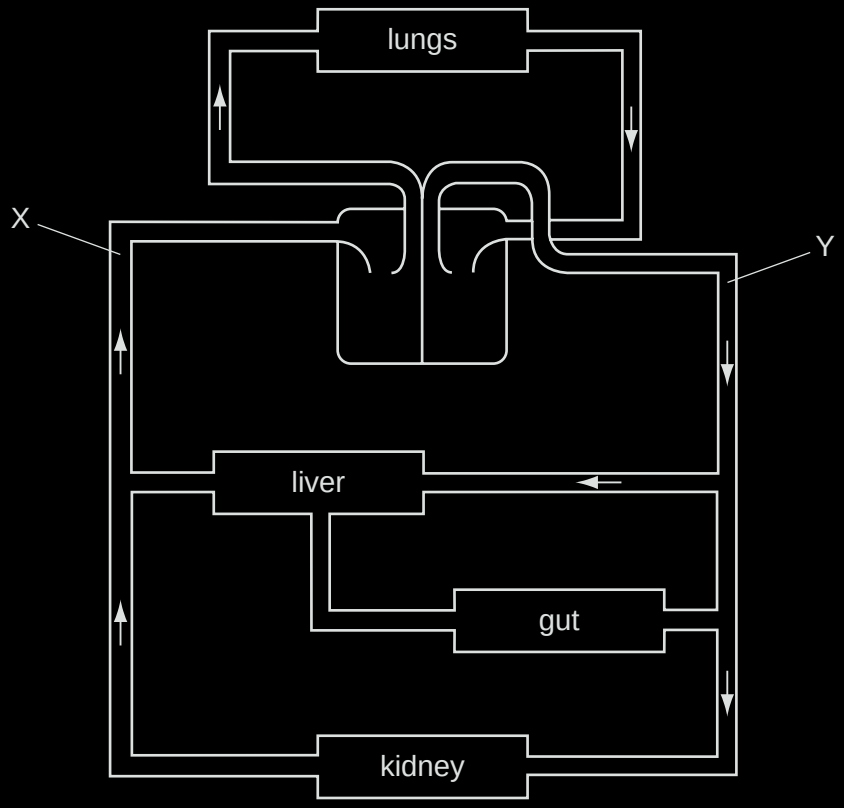
	carbon dioxide concentration	light intensity	temperature
A	1	2	3
B	2	3	1
C	3	1	2
D	3	2	1



After 20 minutes at 35 °C a sample of the water around the bag in each tube is boiled with Benedict's solution.

What are the expected results?

	tube 1	tube 2	tube 3	tube 4
A	blue	orange	blue	orange
B	blue	orange	orange	blue
C	orange	blue	orange	blue
D	orange	orange	blue	orange



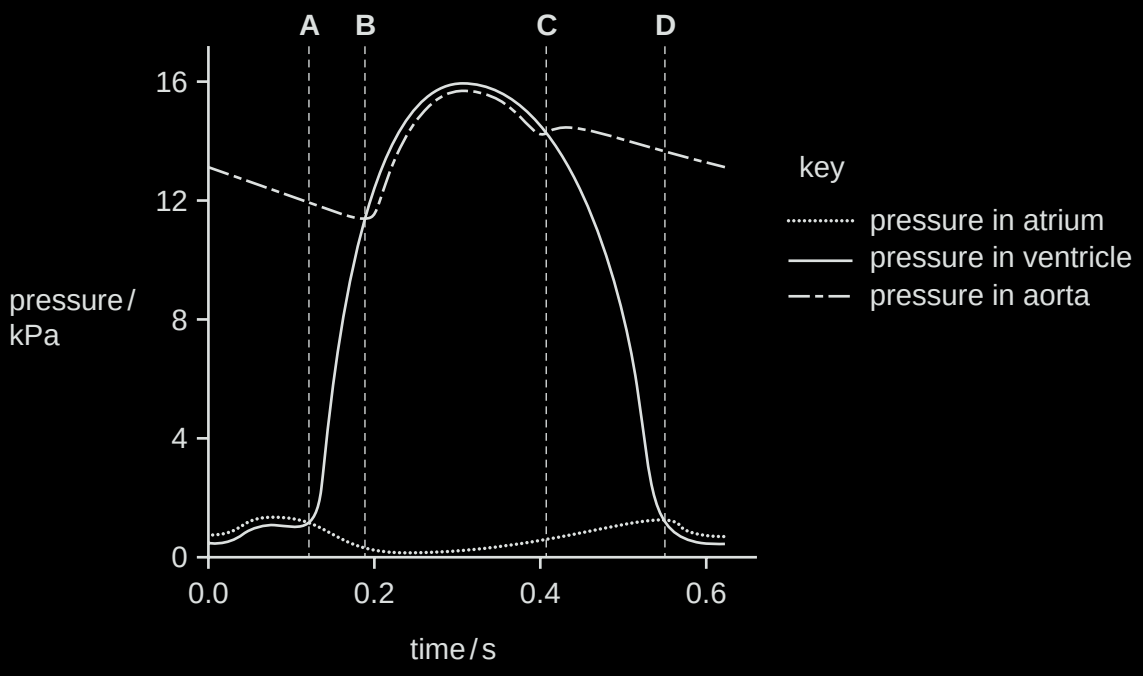
Which vessel **must** the blood pass through in flowing from X to Y?

A hepatic artery

B hepatic portal vein

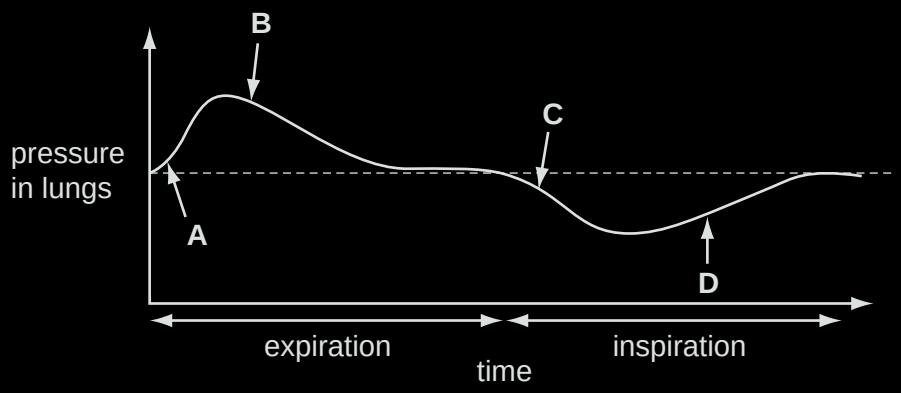
C pulmonary artery

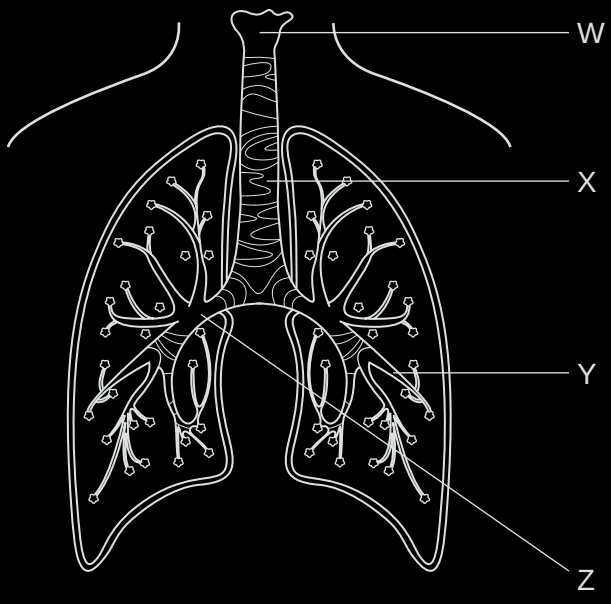
D renal vein



16 The diagram illustrates changes in air pressure taking place inside the lungs during a complete cycle of breathing.

Which position on the graph corresponds to the point at which the ribs are beginning to be raised?



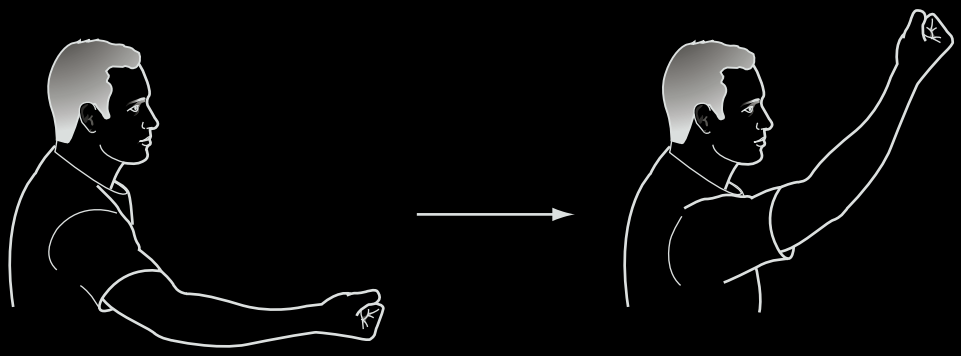


What are W, X, Y and Z?

	bronchus	bronchiole	larynx	trachea
A	W	X	Z	Y
B	X	Z	Y	W
C	Y	W	X	Z
D	Z	Y	W	X

18 When breathing out, which changes occur in the volume of the thorax, the rib cage and the diaphragm?

	volume of thorax	rib cage	diaphragm
A	decreases	lowered	rises
B	decreases	raised	pulled down
C	increases	lowered	rises
D	increases	raised	pulled down



Which joint is working to allow this movement?

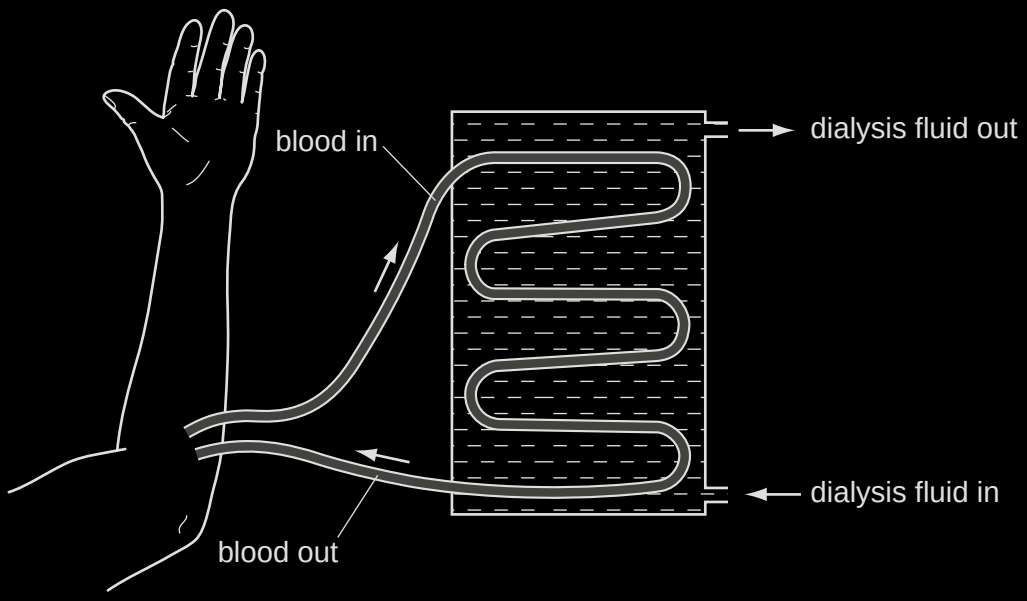
A a ball and socket joint at the elbow

B a ball and socket joint at the shoulder

C a hinge joint at the elbow

D a hinge joint at the shoulder

20 The diagram represents part of a kidney machine.



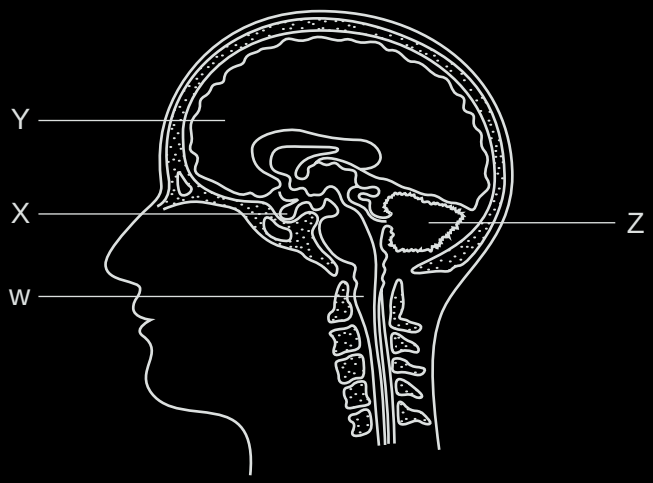
Which substance must be at the same concentration in the dialysis fluid and in the blood?

A glucose

B salt

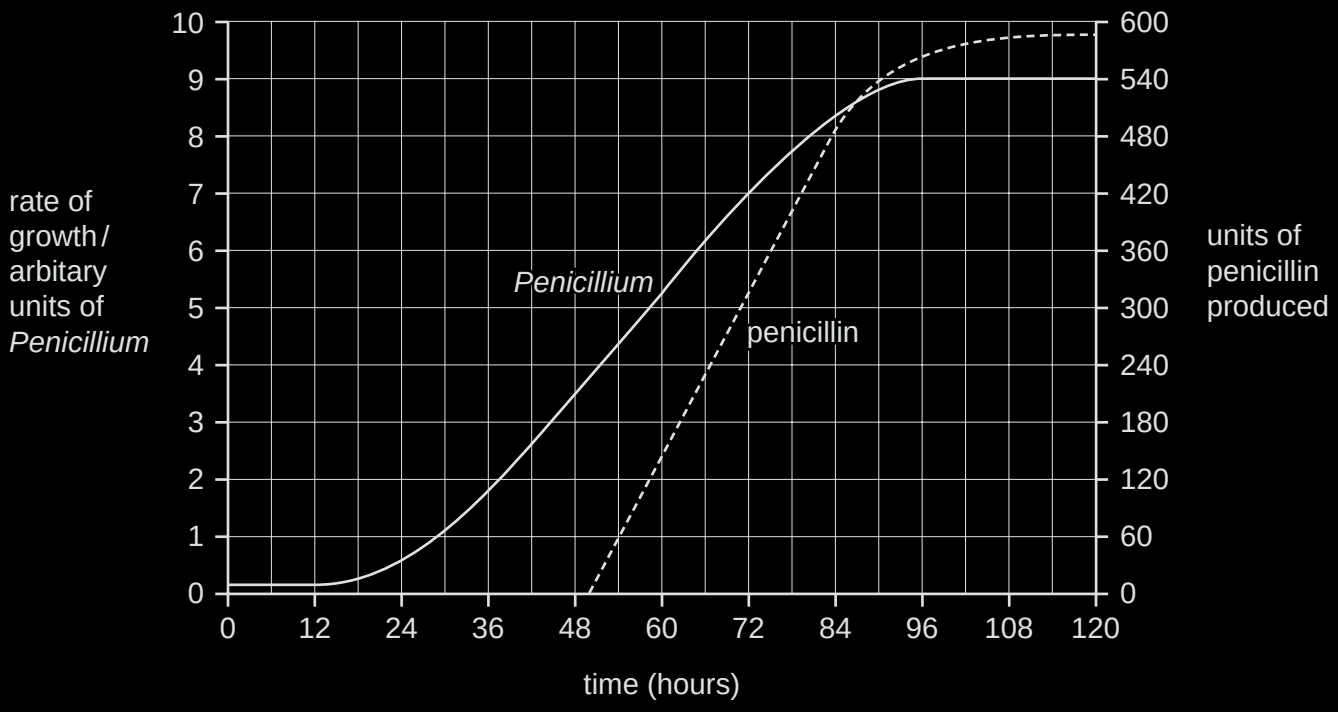
C urea

D water



Where are the cerebellum and the pituitary gland?

	cerebellum	pituitary gland
A	W	Z
B	X	Y
C	Y	W
D	Z	X



After how long should the penicillin be removed from the fermenter to obtain the most yield in the shortest time?

A 54 hours **B** 80 hours **C** 96 hours **D** 120 hours

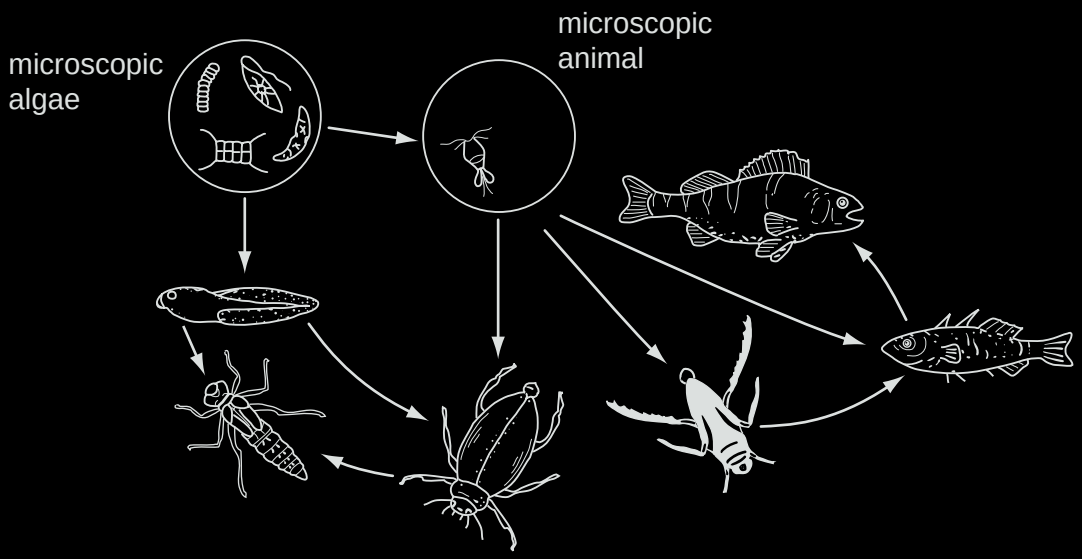
28 Which situation does **not** involve the cycling of the factor named?

A carbon dioxide between plants, animals and the atmosphere

B energy along a food web of producers and consumers

C nitrogen between plants, animals, bacteria and the atmosphere

D water in an ecosystem

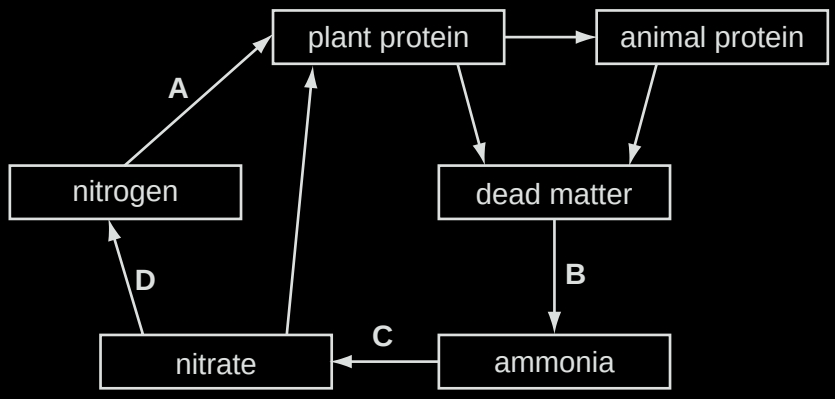


How many primary consumers are shown in this food web?

A 1 **B** 2 **C** 3 **D** 4

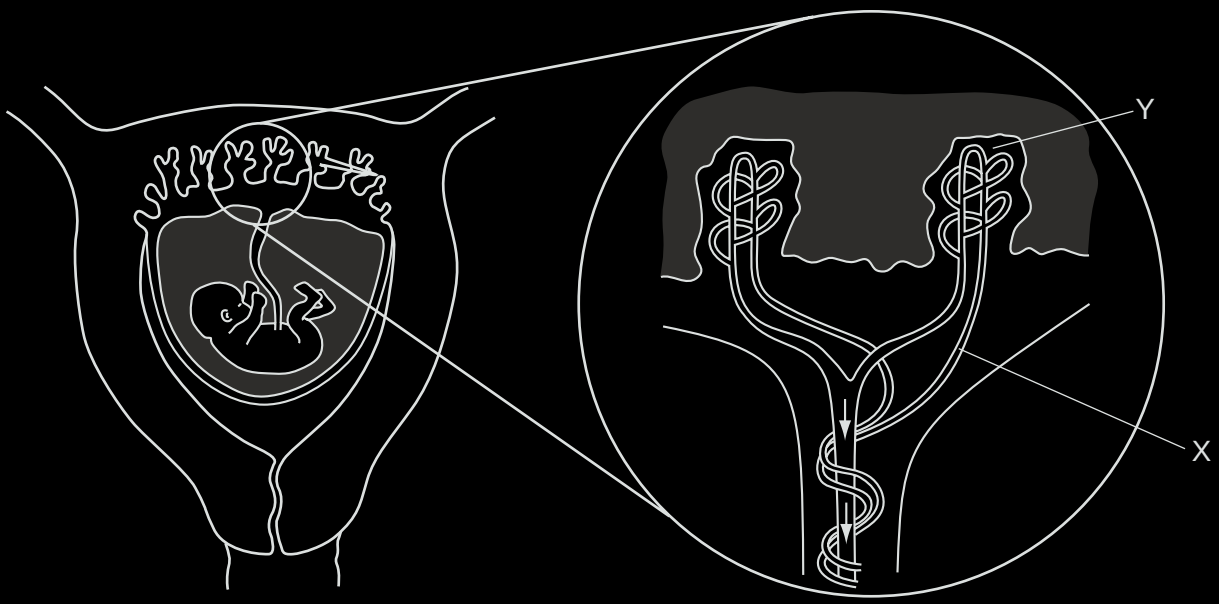
30 The diagram shows part of the nitrogen cycle.

Which stage depends on nitrifying bacteria?



31 What are the roles of the mosquito and the malarial parasite in the spread of malaria?

	mosquito	malarial parasite
A	host	vector
B	pathogen	host
C	pathogen	vector
D	vector	pathogen



Which substances will be at a higher concentration at Y than at X?

A carbon dioxide and glucose

B carbon dioxide and urea

C glucose and oxygen

D oxygen and urea



red form
with black spots



black form
with red spots

What kind of variation is shown by the beetle and why?

A continuous variation because it is controlled by genes

B continuous variation because there are two forms

C discontinuous variation because it is controlled by genes

D discontinuous variation because the two forms are distinct

38 A pregnant woman is told by a genetic counsellor that her baby has equal chances of being blood group A or blood group AB.

What are possible genotypes of the woman and her husband?

A AA and BO

B AB and BO

C AO and BB

D AB and AO

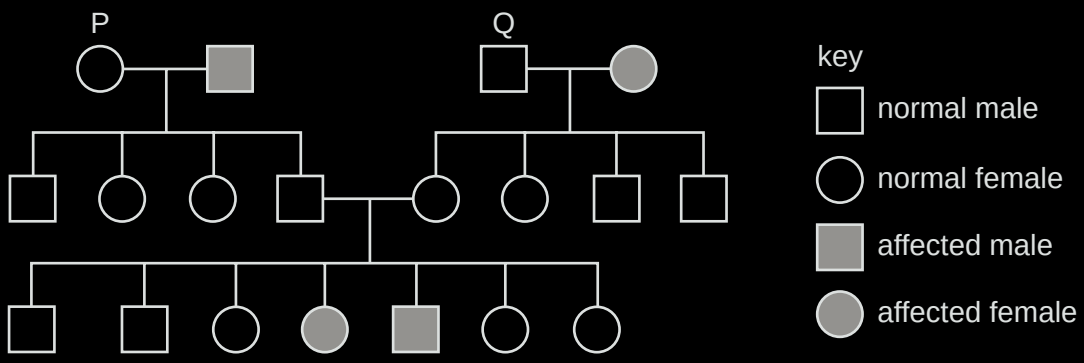
39 Which statement about the genotypes of organisms is correct?

A Dominant alleles are only found in homozygotes.

B One recessive allele always causes a recessive phenotype.

C Recessive phenotypes must be homozygous.

D The dominant phenotype must be heterozygous.



T represents the dominant allele and t represents the recessive allele.

What are the most likely genotypes of individuals P and Q?

	P	Q
A	Tt	Tt
B	Tt	TT
C	TT	Tt
D	TT	TT